

LEARNING TASK 1

Course ID: Probability & Statistics

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The Role of Statistics in Engineering

To illustrate the fundamental importance of probability and statistics in engineering, we start by looking at two practical examples drawn from the bachelor **Military Systems and Technology**.

Example 1.1: Maintenance: reliability of a circuit

Consider a network circuit that functions only if there exists an unbroken path of operational components leading from the left terminal to the right terminal. The system depicted in Figure 1 consists of three distinct components. Devices *A* and *B* are connected in parallel, and Device *C* is connected in series with the parallel pair.

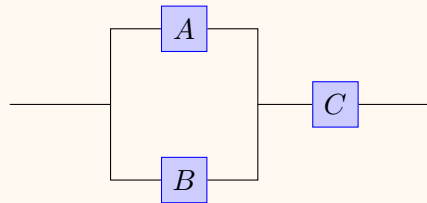


Figure 1: Circuit configuration with three operational devices.

Due to the system geometry, the circuit successfully operates if Device *C* and at least one of Devices *A* or *B* function properly. Conversely, if Device *C* fails, the entire circuit fails regardless of the states of *A* and *B*.

Suppose now that each individual device has an expected average lifetime of 10 hours. We wish to answer the following question: **what is the probability that the overall circuit remains operational for at least 9 hours?**

We analyze this scenario under two different modeling perspectives:

- From a **deterministic perspective**, variability is completely ignored. In other words, all devices are assumed to operate for exactly their expected lifetime of 10 hours. Under this assumption, every component is guaranteed to survive past 9 hours. Consequently, the probability that the system operates for at least 9 hours is equal to 1 (or 100 %).
- From a **stochastic perspective**, we take into account that component lifespans exhibit random fluctuations and variability. When incorporating random variation (modeling lifespans as random variables), the actual probability drops significantly to approximately 0.52 (or 52 %).

This result can be verified through Monte Carlo simulation (using software such as Excel) or calculated analytical methods in probability theory. In a simulation approach, we simulate 10.000 independent trials of the circuit (see “Excel Simulation - Learning Task 1”):

1. For each trial, three individual lifespans are pseudo-randomly generated for *A*, *B*, and *C* according to an appropriate failure distribution.
2. The total lifespan of the system is computed in Column *D* as $\min(\max(A, B), C)$.
3. Column *E* evaluates whether the overall circuit lifespan exceeds 9 hours, recording 1 for success and 0 for failure.

4. Averaging these outcomes over all 10.000 trials yields an estimate of the overall survival probability in Cell **E10004**.

Bestand

Start

Invoegen

Pagina-indeling

Formules

Gegevens

Controleren

Beeld

Geef aan wat u wilt doen...

Van Access

Van web

Van tekst

Van andere bronnen

Bestaande verbindingen

Nieuwe query

Query's weergeven

Uit tabel

Recente bronnen

Alles vernieuwen

Verbindingen

Eigenschappen

Koppelingen bewerken

Sorteren

Externe gegevens ophalen

Ophalen en transformeren

Verbindingen

Sort

E10004

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Figure 2: Simulated lifespan records and empirical success probability of the circuit.

Example 1.2: Operations research: search and detection

In modern search theory, we evaluate defensive barriers deployed to detect incoming threats. Consider an observer patrolling a straight corridor or lane of total width D while moving at a constant patrol speed V following a periodic back-and-forth pattern. This setting is depicted in Figure 3.

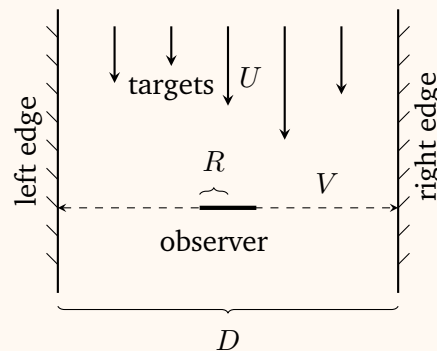


Figure 3: Linear barrier search pattern with reversing courses at lane boundaries.

The observer employs a binary detection sensor with a fixed sweep radius R : any target entering within a distance R of the observer is detected with certainty, whereas targets outside this radius remain completely undetected.

Incoming targets attempt to traverse the corridor moving strictly southwards at a constant velocity U . Assuming we know the capabilities of the target (speed U) but not its entry position, we treat its lateral position as a continuous random variable X .

If a target intentionally avoids the center to minimize detection risk, it might adopt a non-uniform strategy where entry near the boundary edges is favored. Suppose the spatial probability density function $f(x)$ for the target's position across a lane of width $D = 20$ nm (centered at $x = 0$) is given by the V-shaped distribution in Figure 4.

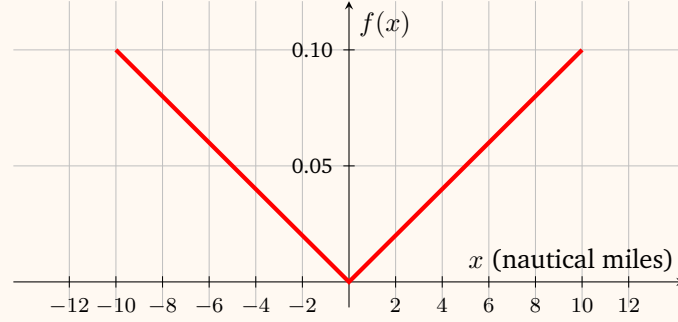


Figure 4: Probability density function $f(x)$ governing the target's lateral position.

Under this specific spatial density, the probability that the target enters within the central region $[-5, 5]$ is given by the area under $f(x)$:

$$P(-5 \leq X \leq 5) = \int_{-5}^5 f(x) dx = 0.25.$$

Conversely, the probability that the target enters along the outer regions $[-10, -5] \cup [5, 10]$ is 0.75.

By integrating probabilistic geometric detection sweep coverage across time, the overall probability of detecting a target (P_{det}) can be derived analytically as:

$$P_{\text{det}} = \left\{ \frac{4R^3}{3D^3} \left(1 + \left(\frac{V}{U} \right)^2 \right) - \frac{2R^2}{D^2} \sqrt{1 + \left(\frac{V}{U} \right)^2} + \frac{2R}{D} \right\} \sqrt{1 + \left(\frac{V}{U} \right)^2}$$

Example Calculation Given parameters $D = 20$ nm, $R = 4$ nm, observer speed $V = 18$ knots, and target speed $U = 12$ knots:

$$P_{\text{det}} \approx 0.5236 \quad (52.36 \%).$$

Probability Concepts and Axioms

We start by formalizing a concept that can be used to study random phenomena.

Definition 1.3: Probability model

A *probability model* is characterized through the following three properties:

- **Random Experiment:** a process or procedure that leads to one of several possible outcomes, where the precise outcome cannot be predicted with certainty in advance.
- **Sample Space (S):** the set of all possible outcomes of a random experiment.
- **Event ($A, B, \dots$):** any subset of outcomes contained within the sample space S .

Example 1.4: Target engagement

Consider an engagement model where a military weapon system engages a target iteratively until the target is successfully eliminated.

- Let h represent a hit (target destroyed) and m represent a miss.
- The sample space is infinite and countable:

$$S = \{h, mh, mmh, mmmh, \dots\}$$

- Define Event A as “eliminating the target in no more than 3 attempts”:

$$A = \{h, mh, mmh\} \subset S$$

- Define Event B as "requiring at least 2 attempts to eliminate the target":

$$B = \{mh, mmh, mmmh, \dots\} \subset S$$

Set Operations on Events

- **Complement (A' or A^c):** The event consisting of all outcomes in S that are not in A (not- A).
- **Intersection ($A \cap B$):** The event containing outcomes that belong to *both* A and B .
- **Union ($A \cup B$):** The event containing outcomes that belong to A , or B , or both.
- **Mutually exclusive (disjoint):** Two events A and B are mutually exclusive if they share no common outcomes, meaning $A \cap B = \emptyset$.

Applying these set operations to our engagement model:

$B' = \{h\}$	The target is eliminated on the very first attempt.
$A \cap B = \{mh, mmh\}$	The target is eliminated on exactly the 2nd or 3rd attempt.
$A \cup B = S$	Guaranteed outcome (the certain event).

Definition 1.5: Axioms of probability

Let S be a sample space associated with a random experiment. A probability function P is a real-valued function that assigns to each event $E \subset S$ a real number $P(E)$, satisfying the three formal Axioms of Probability:

(A1) Certain Event: $P(S) = 1$ (the probability of the entire sample space is 1).

(A2) Non-negativity: $0 \leq P(E) \leq 1$ for any event $E \subset S$.

(A3) Mutual Exclusivity: If E_1 and E_2 are pairwise mutually exclusive events ($E_1 \cap E_2 = \emptyset$), then:

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Core Probability Rules

From the foundational axioms, we derive the fundamental algebraic rules of probability theory:

(R1) Impossible Event: $P(\emptyset) = 0$.

(R2) Complement Rule: $P(A') = 1 - P(A)$.

(R3.1) General Addition Rule (Two Events):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

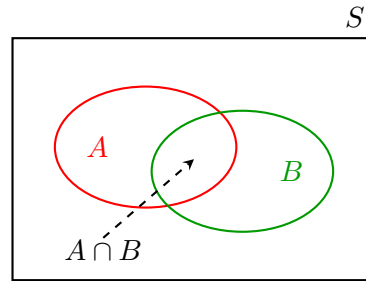


Figure 5: If you compute the probability $P(A \cup B)$, adding the probabilities $P(A)$ and $P(B)$ means you have counted the probability for the intersection $P(A \cap B)$ twice.

Proof of (R3.1)

Using set decomposition, we express $A \cup B$ and B as unions of disjoint sets:

$$A \cup B = A \cup (B \cap A') \quad \text{and} \quad B = (A \cap B) \cup (B \cap A')$$

Applying Axiom (A3) to both disjoint decompositions gives:

$$P(A \cup B) = P(A) + P(B \cap A')$$

$$P(B) = P(A \cap B) + P(B \cap A') \implies P(B \cap A') = P(B) - P(A \cap B)$$

Substituting $P(B \cap A')$ back into the first expression yields the General Addition Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

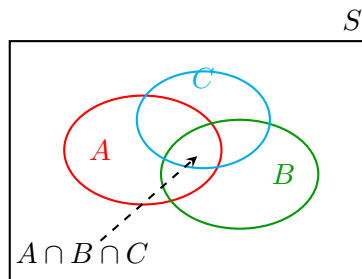


Figure 6: Venn diagram for the union of three overlapping events A , B , and C .

(R3.2) Addition Rule for Three Events:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Exercise

Consider rolling two fair six-sided dice. The sample space S consists of 36 equally likely ordered pairs (i, j) for $i, j \in \{1, \dots, 6\}$.

- Let $A = \text{"sum is at least 10"} = \{(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$.
- Let $B = \text{"sum is no more than 10"}.$

Calculating the corresponding probabilities $P(A)$, $P(B)$, $P(A \cap B)$ and $P(A \cup B)$.

Solution

$$P(A) = \frac{|A|}{36} = \frac{6}{36} = \frac{1}{6}$$

$$P(B) = 1 - P(B') = 1 - P(\text{sum is 11 or 12}) = 1 - \frac{3}{36} = \frac{33}{36} = \frac{11}{12}$$

$$P(A \cap B) = P(\text{sum is exactly 10}) = P(\{(4, 6), (5, 5), (6, 4)\}) = \frac{3}{36} = \frac{1}{12}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{6}{36} + \frac{33}{36} - \frac{3}{36} = \frac{36}{36} = 1$$

Conditional Probability

Conditional probability measures the probability of an event B occurring given that another event A has already occurred.

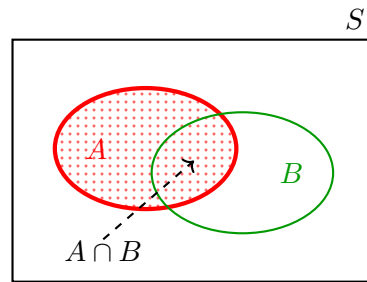


Figure 7: Reduced sample space when conditioning on the occurrence of event A .

Definition 1.6: conditional probability

Formally, for any event A such that $P(A) > 0$, the *conditional probability* of B given A is defined as:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication and Total Probability Rules

Rearranging the definition of conditional probability provides the **Multiplication Rules**:

$$(R4.1) \quad P(A \cap B) = P(A) \cdot P(B|A)$$

$$(R4.2) \quad P(A \cap B) = P(B) \cdot P(A|B)$$

Total Probability Rule (Two-Event Form) For any event B and condition event A :

$$(R5) \quad P(B) = P(B|A)P(A) + P(B|A')P(A')$$

Proof of (R5)

Decomposing B into mutually exclusive components relative to A :

$$B = (B \cap A) \cup (B \cap A')$$

Taking probabilities using Axiom (A.3) and applying Rule (R4.2) gives:

$$P(B) = P(B \cap A) + P(B \cap A') = P(B|A)P(A) + P(B|A')P(A')$$

Example 1.7: semiconductor quality control

Suppose that in semiconductor manufacturing, the probability is 0.10 that a chip subjected to high levels of contamination during manufacturing causes a product failure. The probability is 0.005 that a chip not subjected to high contamination levels during manufacturing causes a product failure. In a particular production run, 20 % of the chips are subject to high levels of contamination. What is the probability that a product using one of these chips fails?

Solution

It helps to look at this problem by defining relevant events that we can apply the probability rules to, given the information:

Let A be the event “chip is subjected to high levels of contamination”, and B the event “semiconductor fails”. We know the following probabilities already:

- Probability of failure given high contamination: $P(B|A) = 0.10$.
- Probability of failure given normal conditions: $P(B|A') = 0.005$.
- Proportion of chips subjected to high contamination: $P(A) = 0.20$ (thus $P(A') = 0.80$).

We determine the total probability of chip failure $P(B)$ using a probability tree diagram:

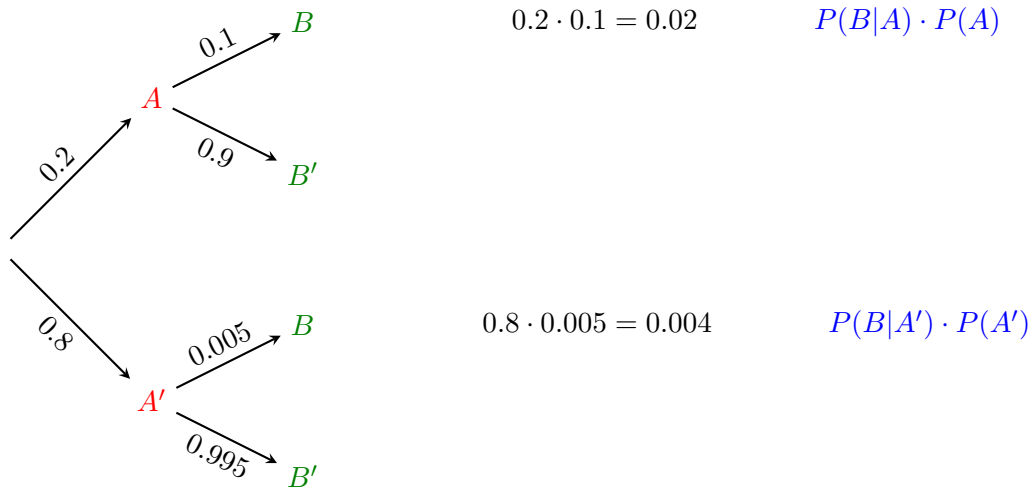


Figure 8: Probability tree structure for semiconductor contamination and failure analysis.

Summing the path probabilities yields:

$$P(B) = P(B|A) \cdot P(A) + P(B|A') \cdot P(A') = 0.020 + 0.004 = \mathbf{0.024} \quad (2.4 \%)$$

General Law of Total Probability If the sample space S is partitioned into k mutually exclusive and exhaustive events $A_1, A_2, \dots, A_k$ (such that $\bigcup_{i=1}^k A_i = S$ and $A_i \cap A_j = \emptyset$ for $i \neq j$), then for any event B :

$$P(B) = \sum_{i=1}^k P(B|A_i) \cdot P(A_i)$$